

B. Tech 1st Semester Sessional Examination-I
Subject-Mathematics-II
Branch- CE (Sec-C), Total Marks-30, Date- 17-02-2025

Answer the following questions:

1. Solve (any three):

5x3=15

a) $\frac{dy}{dx} - 2xy = e^{x^2}$

b) $x \frac{dy}{dx} = y - \cos \frac{1}{x}$

c) $\frac{dy}{dx} + \frac{y}{x} = \frac{y^3}{x^2}$

d) $(y \cos x + 1)dx + \sin x dy = 0$

e) $(x^2 + y^2 - a^2)xdx + (x^2 - y^2 - b^2)ydy = 0$

2. Answer the following:

5x3=15

a) Check whether the function $f(z) = 2z^2 - 3iz$ is analytic or not.

b) Show that the function $u = x^3 - 3xy^2$ is harmonic

c) Determine the analytic function $f(z)$ whose real part is
 $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$